

Polymer Dynamics

Static Structure Factor Analysis

Weekly Progress Report 4

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1 Introduction

This week focused on the structural characterization of the polymer backbone during thermal annealing through the calculation of the static structure factor.

The structure factor provides information about spatial correlations in the polymer chain and enables characterization of structural organization across different length scales. By studying the evolution of the structure factor across different annealing blocks, changes in polymer conformation during cooling can be quantified.

2 Objectives

1. Compute the static structure factor of the polymer backbone.
2. Analyze structural evolution during annealing.
3. Compare scattering profiles across annealing blocks.
4. Identify scaling regimes in the structure factor.
5. Extract power-law exponents from log-log plots.
6. Investigate low- k and high- k polymer behavior.

3 Simulation Details

The supplied trajectory file was

Nb100Ns20gd1.0.lmptrj

Only backbone monomers (atom type 1) were included in the analysis.

The simulation box dimensions were

$$L_x = 60, \quad L_y = 40, \quad L_z = 150.$$

The trajectory was divided into four annealing blocks:

Block	Timestep Range
1	$0 - 2.5 \times 10^7$
2	$2.5 \times 10^7 - 5.0 \times 10^7$
3	$5.0 \times 10^7 - 7.5 \times 10^7$
4	$7.5 \times 10^7 - 1.0 \times 10^8$

Table 1: Annealing blocks used for structure factor analysis

A total of 20 sampled configurations were analyzed for each annealing block after frame skipping and blockwise averaging.

4 Static Structure Factor

The static structure factor was calculated using

$$S(\mathbf{k}) = \frac{1}{N} \left\langle \left| \sum_{j=1}^N e^{i\mathbf{k} \cdot \mathbf{r}_j} \right|^2 \right\rangle$$

- N is the number of backbone monomers,
- $\mathbf{r}_j$ is the position of monomer j ,
- averaging is performed over sampled configurations.

Wave vectors satisfying periodic boundary conditions were generated using

$$k_x = \frac{2\pi n_x}{L_x}, \quad k_y = \frac{2\pi n_y}{L_y}, \quad k_z = \frac{2\pi n_z}{L_z}.$$

To obtain an isotropic scattering profile, shell averaging was performed by grouping wave vectors with similar magnitudes and averaging their corresponding scattering intensities.

4.1 Power-Law Scaling

Scaling behavior was analyzed by fitting linear regions of the log-log structure factor curves using

$$S(k) = Ak^{-\alpha},$$

- A is the fitting prefactor,
- α is the scaling exponent.

The exponent α characterizes the conformational properties of the polymer over different length scales.

4.2 Extracted Exponents

Block	k Range	Exponent α
3	$0.10 < k < 0.30$	1.564
3	$0.80 < k < 2.50$	1.410
4	$0.10 < k < 0.30$	1.478
4	$0.80 < k < 2.50$	1.335

Table 2: Power-law exponents obtained from structure factor fitting

4.3 Structure Factor Plot

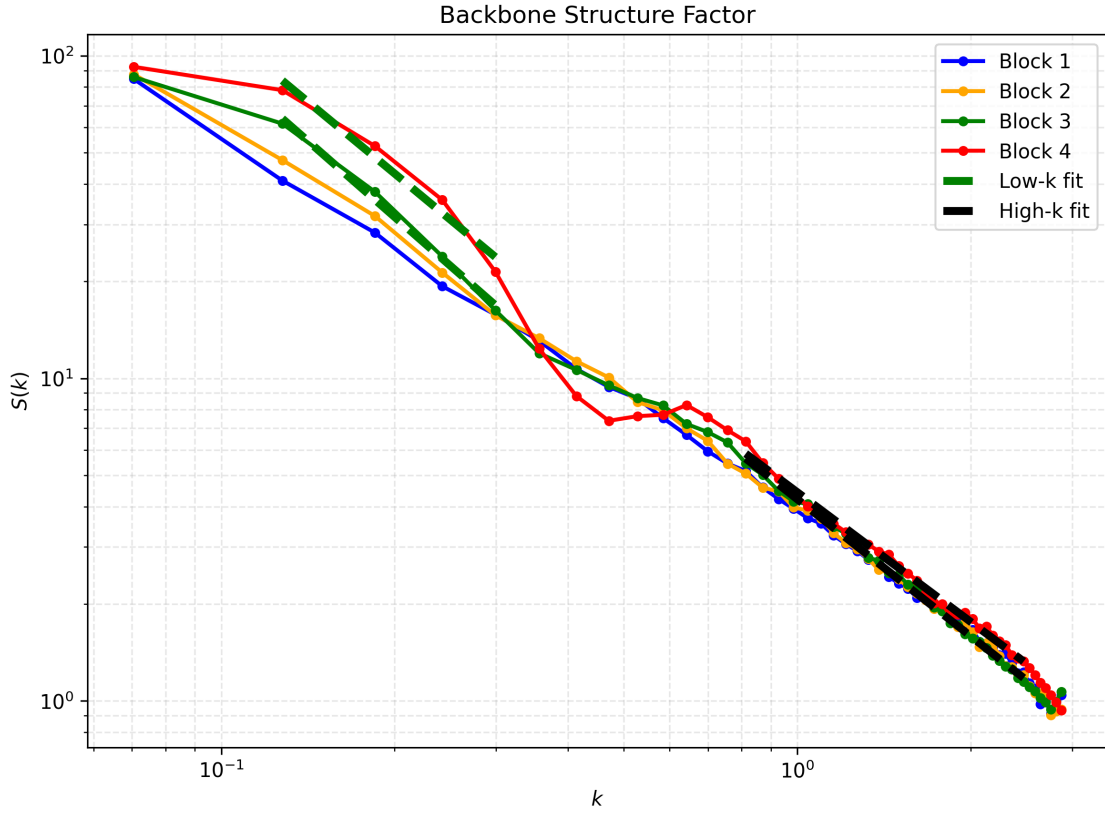


Figure 1: Blockwise static structure factor of the polymer backbone during annealing. Dashed lines indicate power-law fits performed in the scaling regimes.

4.4 Discussion

The following observations were made:

- All blocks show a strong increase in $S(k)$ at small wave vectors.
- Structural differences are most pronounced in the low- k regime.
- At high wave vectors, all curves approach $S(k) \approx 1$.

- Block 3 and Block 4 exhibit clear scaling regions suitable for power-law fitting.
- A crossover region is visible in Block 4 at intermediate wave vectors.
- The crossover region does not exhibit linear behavior and was therefore not fitted.
- The extracted exponents lie between approximately 1.3 and 1.6.

Following mentor feedback, an attempt was made to fit the very low- k regime ($k < 0.1$). However, after shell averaging, the lowest available shell-averaged wave vector was found to be

$$k \approx 0.128.$$

Consequently, no shell-averaged data points were available below $k = 0.1$, preventing a separate power-law fit in that region.

5 Conclusion

Blockwise structure factor analysis of the polymer backbone was successfully performed during the annealing process.

Distinct structural differences were observed between annealing blocks, particularly in the low-wave-vector regime. Power-law exponents were extracted from the accessible scaling regions and were found to vary between 1.3 and 1.6.

The analysis revealed a crossover region in Block 4 and provided quantitative insight into the structural evolution of the polymer during annealing. These results complement previous MSD and Radius of Gyration analyses and contribute to a more comprehensive understanding of polymer dynamics.

6 Software and Libraries Used

- Python
- NumPy
- Matplotlib
- LAMMPS trajectory files (.lammprj)
- Fourier-based structure factor calculation
- Log-log power-law fitting